ANIKET ANAND

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EDUCATION

The University of Chicago *expected 2028*
Ph.D. in Computer Science

Georgia Institute of Technology *May 2023*
M.S. in Computer Science *GPA: 3.91/4*

Indian Institute of Technology (BHU) *Jul 2020*
B.Tech in Ceramic Engineering *CGPA: 9.34/10*

RESEARCH INTERESTS

Understanding the safety risks of capable LLMs, application of ML methods for security and networks problems

PAPERS

- **A. Anand***, J. Singh*, Z. Sun, D. Hakkani-Tür, and N. Feamster (*equal contribution). Measuring, Localizing, and Ablating Alignment Signatures in LLMs. Under review, 2026.
- **A. Anand**, Y. Hou, D. Fields, A. Kantchelian, D. Tao, K. Thomas, and G. Ho. Benchmarking and Exploring the Capabilities of LLMs for Attack Investigations. Under review, 2026.
- N. Kostyuk, J. Lindsay, E. E. Kim, **A. Anand**, Z. Bischof, A. Meng, and A. Dainotti. Strategic Interdependence: Using Internet Outage Data to Study How Combatants Manage Collective Institutions During War. British Journal of Political Science, 56 (2026), e17
- **A. Anand**, Z. Bischof, A. Meng, and A. Dainotti. Tracking Internet connectivity disruptions during prolonged military conflict: the case study of the Russia - Ukraine war 2024.
- **A. Anand**, M. Kallitsis, J. Sippe, and A. Dainotti.Aggressive Internet-Wide Scanners: Network Impact and Longitudinal Characterization. CoNEXT 2023.
- **A, Anand**, W. Asif, and M. Lestas. Performance Evaluation of PoW Blockchain in Wireless Mobile IoT networks. In 2021 17th International Conference on Distributed Computing in Sensor Systems (DCOSS), pp. 396-403. IEEE, 2021.
- **A. Anand**, A. Galletta, A. Celesti, M. Fazio, and M. Villari. A secure inter-domain communication for IoT devices. In 2019 IEEE International Conference on Cloud Engineering (IC2E), pp. 235-240. IEEE, 2019.
- S. Gupta, R.S. Singh, and **A. Anand**. Cloudlet Scheduling using Merged CSO algorithm. In 2018 Fifth International Conference on Parallel, Distributed and Grid Computing (PDGC), pp. 278-283. IEEE, 2018.

KEY RESEARCH EXPERIENCES

Understanding the AI detectability gap between Base and Aligned Language models for generating natural-sounding text *UChicago: Winter 2026*

- Showed that post-training imparts a detector-visible AI style: under matched prefixes, aligned-model generations trigger far higher AI-detection rates than their base counterparts (e.g., 98.5% vs. 9.2% on Pangram) and drift away from human-corpus text.
- Introduced PASTA (Post-training Alignment Signature Targeted Ablation), which localizes this style to a residual-stream direction via aligned–base activation contrasts and ablates it during decoding, requiring only 25 calibration examples and no parameter updates.

- Demonstrated that ablating this single direction cuts detection rates by 73–98 percentage points on Pangram for 8 of 11 aligned models (evaluated against 6 detectors), while generations remain relevant and coherent with greater stylistic variation.

Predictability analysis of enterprise-grade resource access logs *Google: Summer 2025*
Dr. Alex Kantchelian

- Characterized the predictability of the CERT data using a high-precision ML-based anomalous access detection system called Facade. Also contributed to open-source version of Facade.
- Trained Facade's model on the context and action features extracted from CERT's machine login traces, file, email access traces, after fine-tuned the hyperparameters of model using Vizier, and inferred anomalous scores of malicious accesses.
- Discovered very regular and periodic access patterns in the CERT dataset, and found that anomalous events are clustered together in the CERT dataset. This showed a need for more realistic access traces.

Feasibility of ML for anomalous network packet detection *Merit Network: Summer 2023*
Dr. Michalis Kallitsis

- Analyzed the network PCAPs received unsolicited at a large unused block of IPv4 address space. Identified malicious PCAPs by using Suricata's ground truth labels.
- Explored various ML methods for evaluating security posture of the Internet, such as detection of DoS while taking into consideration the concept drift, based on the suspicious PCAPs.

Internet connectivity in Ukraine following the 2022 invasion *M.S. research: Spring 2023*
Prof. Alberto Dainotti

- Characterized Internet disruptions in Ukraine since the beginning of the Russian invasion using Internet Outage Detection and Analysis (IODA's) active probing signals and created a ground truth dataset of drops in the Internet connectivity.
- Developed a methodology for improving automated detection of Internet outages by comparing a time-series forecasting model's predicted connectivity level with the observed level.

Aggressive Internet-wide scanners and their network impact *Merit Network: Summer 2022*
Dr. Michalis Kallitsis and Prof. Alberto Dainotti

- Identified aggressive internet scanners by leveraging data from a large Network Telescope and found that the aggregate impact of such scanners can be as high as 4.7% of total packets processed by the largest border router of an ISP serving over 1 million customers on a typical day
- Performed longitudinal analysis of these aggressive scanners with approximately 2 years of data and characterized targeted ports, fingerprintable tools used for scanning, scan origin, rate of scan, etc.d

Taxonomize major Internet networks *M.S. research: Spring 2022*
Prof. Alberto Dainotti

- Identified Autonomous System Numbers (ASNs) with large footprint in the U.S. and Ukraine by leveraging two data sources namely: 1. a combination of prefix to AS mapping from BGP data and MaxMind IP-geolocation data and 2. APNIC Eyeballs dataset
- Manually classified over 1000 major ASes into 6 types such as Residential ISP, Educational Institutes, etc. by leveraging RIR WHOIS records, PeeringDB, and organization websites as main data sources

TEACHING ASSISTANT EXPERIENCE

- CMSC 14300: Systems Programming I, University of Chicago *Spring 2026*
- CMSC 33250: Graduate Computer Security, University of Chicago *Fall 2025*
- CS 6601: Artificial Intelligence, Georgia Institute of Technology *Spring 2022*

RELEVANT COURSES UNDERTAKEN

Grad-level: Natural Language Processing, Machine Learning, Artificial Intelligence, Using NLP and LLMs for Computer Security, AI/ML for Computer Security, Internet Data Science
Undergrad: Network Security, Natural Language Processing, Operations Research, Computer Programming, Mathematical Methods

SCHOLASTIC ACHIEVEMENTS AND ACCOLADES

- Nominated for Sahaj Memorial Award of AIPMA for best student in Ceramic Engineering (2020)
- Selected among 35 students from India for NTU-India Connect Research Internship (2019)
- Secured All India Rank within top 4% in JEE-Advanced out of 200,000+ aspirants (2016)
- Achieved All India Rank within top 0.8% in JEE-Main out of 1.2 million+ aspirants (2016)
- Secured a top 300 spot in National Standard Examination in Junior Science (NSEJS) (2013)